

# Path Dependency and Cement Industry Decarbonisation in the UK: A Case Study of the Peak Cluster CCS Project

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The cement industry accounts for approximately seven to eight per cent of global anthropogenic CO<sub>2</sub> emissions, making it one of the most challenging sectors to decarbonise. In the United Kingdom, the Peak Cluster–Morecambe project proposes a ~200 km onshore CO<sub>2</sub> pipeline from four anchor cement and lime facilities in Derbyshire and Staffordshire to depleted offshore gas reservoirs in the Irish Sea. This thesis evaluates policy drivers, economic viability and public perception through a convergent parallel mixed-methods design.

Analysis of fifteen UK government and industry documents, using Erickson et al. lock-in indicators and Bacchi's problem-representation framework, finds a self-reinforcing institutional architecture (mean path-dependency 4.4 / 7). Sunset and pathway-diversification clauses appear in only 13 per cent of instruments.

A Monte Carlo simulation of 10,000 iterations finds that the median levelised cost of CO<sub>2</sub> avoided for post-combustion MEA is USD 131.4 per tonne, against a 2026 UK ETS civil-penalty reference of USD 65.2 per tonne — implying on the order of USD 199 million per year of public support at the project's three-megatonne design throughput. Comparative MAC analysis shows that clinker substitution and related options can deliver meaningful abatement at substantially lower cost, while remaining insufficient on their own for process emissions.

Discourse analysis of 191 public records finds that the project has not secured social licence above conditional withholding: 49 per cent of non-proponent items are coded at withholding and 51 per cent at conditional scrutiny. Dominant frames are place identity and landscape (49 per cent), procedure (34 per cent), and safety and ecological risk (15 per cent).

The joint display identifies a generative mechanism: an entrenched policy regime that financialises a hard-to-abate pathway can simultaneously produce the legitimacy deficit that becomes consequential at Development Consent Order. Industrial decarbonisation in the UK is path-dependent, but it is not deterministic.

**Keywords.** *carbon capture and storage; carbon lock-in; path dependency; cement decarbonisation; social licence to operate; techno-economic analysis; Peak Cluster; UK industrial policy*